* M decides a language A
if given x
accept if $x \in A$
reject if $x \notin A$
* a decider always halts
* M recognizes A
if $A = L(M)$
* A is mapping reducible to B ($\boxed{A \le_m B}$) if $\exists$ computable $f: \Sigma^* \to \Sigma^*$ s.t. $x \in A \Leftrightarrow f(x) \in B$

$\left.\begin{array}{l} A \le_m B \\ A \text{ is undecidable} \end{array}\right) \Rightarrow$ so is B

* undecidable languages:
L_{ACC}. L_{HALT}. L_{EQIV}
undecidable $\Rightarrow$ NP-hard
P efficiently decidable
e.g. GCD, Euclidean path

NP efficiently verifiable
$\exists$ efficient V s.t.
accept x given c if $x \in L$ for some c
reject x given c if $x \notin L$ for any c
e.g. $L_{TSP} = \{(G,k) \mid \exists$ TSP with weight $\le k\}$
$L_{\le DIST} = \{(G,d) \mid$ code distance $\le d\}$
$L_{=DIST}$ is not known to be in NP.
* A is polynomial time mapping reducible to B if $\exists$ efficient $f: \Sigma^* \to \Sigma^*$ s.t. $x \in A \Leftrightarrow f(x) \in B$
$\boxed{A \le_p B} \Rightarrow A \le_m B$
$NP \subseteq EXP \Rightarrow$ decidable

$\left.\begin{array}{l} A \le_p B \\ B \in P \end{array}\right) \Rightarrow A \in P$

NP-hard if for every $L \in NP$: $L \le_p A$.
NP-complete if NP & NP-hard
* An efficient alg for any NP-hard prob implies P = NP
e.g. SAT. 3SAT = $\{\Phi \mid \Phi$ is a satisfiable 3CNF formula$\}$
Vertex Cover, Set Cover
3 coloring, best schedule, clique, Max-cut
Ham path. TSP
Independent set.
subset sum

Search Prob

* max: max-clique, knapsack
* min: min vertex cover, TSP
* exact: sat, hampath.
* A search prob has an efficient algorithm iff its decision ver has

Def α-approximation if

* max: $\alpha OPT \le x \le OPT$ ($\alpha < 1$)
* min: $OPT \le x \le \alpha OPT$ ($\alpha > 1$)

e.g. 1/2 Knapsack

2 minVC. 1/(Δ+1) max is

2 TSP. 1/2 Max cut (local & rand)

7/8 Max-3sat

* ∃ polinomial verifier ⇒ NP
* NP + NP-hard = NP-complete
* clique k ⇔ vertex cover C

size k |V|-k

edge not in k ⇔ edge covered

* clique k ⇔ independent set S

G compliment graph

edge not in k ⇔ edge in S

* independent set ALLOC

edge in S ⇒ not connected

Diffie-Hellman

Alice $\xrightarrow{x=g^a}$ Bob (key $=g^{ab}$)
$\xleftarrow{y=g^b}$

generator g: $\forall x \neq 0 \in Z_p$
$\exists i \in \mathbb{N}$ s.t. $g^i = x$

* DLOG: Find i s.t. $g^i = x$
$O(\sqrt{p}\log p)$

RSA

Alice $\xrightarrow{e}$ Bob
$\xleftarrow{c = m^e \bmod n}$

key pair (e,d) $ed = 1 \bmod \phi(n)$

Fermat's Little Theorem

① $a^{p-1} = 1 \bmod p$

② $a^p = a \bmod p$

③ $n = pq$ $(p \neq q)$ $\phi = (p-1)(q-1)$
$a^{k\phi(n)+1} = a \bmod n$

$\Rightarrow m' = c^d = \cdots = m \bmod n$

* Difficulty of factoring

(k,n) secret sharing

$S \in Z_q$. random

$f(y) = S + a_1 y + \cdots + a_{k-1} y^{k-1}$

With k points, recover

Reed Solomon

$RS[n,k]_q$

$x = (x_0, \cdots, x_k)$

$f(y) = x_0 + \cdots + x_{k-1} y^{k-1}$

$C(x) = (f_x(1), f_x(2) \cdots f_x(n))$

dist $= n-k+1$

$L_i(y) = \dfrac{\prod_{j \neq i}(y - \alpha_j)}{\prod_{j \neq i}(\alpha_i - \alpha_j)}$

$f(y) = v_1 L_1(y) + \cdots + v_k L_k(y)$

Interactive Proof

isomorphism $f: V_1 \to V_2$
s.t. $(u,v) \in E_1 \Leftrightarrow (f(u), f(v)) \in E_2$

$G_1 \cong G_2$

Fermat's Prime Test

$n \in \mathbb{N}^*$. $a \in [1,n) \cap \mathbb{N}$ witness.

* If p is prime, $a^{n-1} = 1 \bmod n$ for any a.

* Else, $a^{n-1} = 1 \bmod n$ for at most $1/2\, a$.

Markov $Pr[X \geq a] \leq \dfrac{EX[X]}{a}$

Chernoff $X = \sum x_i$; x_i is idd indicator random v.

$EX[x_i] = Pr[x_i = 1] = p$

$Pr[X/n \geq p + \varepsilon] \leq e^{-2\varepsilon^2 n}$

$Pr[X/n \leq p - \varepsilon] \leq e^{-2\varepsilon^2 n}$

$Pr[|X/n - p| \geq \varepsilon] \leq 2e^{-2\varepsilon^2 n}$

Union $Pr[A_1 \cup \cdots \cup A_n] \leq \sum_{i=1}^{n} A_i$

e.g. Extended Euclidean

Find $34^{-1} \bmod 61$

$61 = 34 + 27$
$34 = 27 + 7$
$27 = 3 \times 7 + 6$
$7 = 6 + 1$

$1 = 7 - 6 = 7 - (27 - 3 \times 7)$
$= 4 \times 7 - 27 = \cdots$
$= 9 \times 34 - 5 \times 61$
$\Rightarrow 34^{-1} = 9 \bmod 61$

e.g. Markov

hash table size n^2.

Find prob at least 1 collision

$Ex[Z] = \sum P[h(i) = h(j)] = \binom{n}{2} \frac{1}{n^2} < \frac{1}{2}$

$P[Z \geq 1] \leq P[Z \geq 2 \cdot Ex[Z]] \leq \frac{1}{2}$

e.g. Chernoff 1

A: $v \pm 2\%$. prob $\geq 80\%$. Find B prob $\geq 99\%$

$Pr[Y \leq t/2] = Pr[Y/t \leq \frac{1}{2}] = Pr[Y/t \leq 0.8 - 0.3]$

$\leq e^{-2 \cdot 0.3^2 n} \leq 0.01 \Rightarrow n \geq \cdots$

e.g. Chernoff 2

Approx $1/\pi$ with error < 0.01 with prob > 0.8

Y: # darts in □

$Ex[Y/n] = \frac{2}{\pi}$.

we want $|Y/n - 2/\pi| < 0.02$

$Pr[|Y/n - 2/\pi| \geq 0.02]$

$\leq 2e^{-2 \cdot 0.02^2 n} = 0.2$

e.g. Union

Flip coin n times. Find prob $\exists$ sequence of $(\log_2 n) + k$ consequtive heads

$p \leq (n - \log n - k + 1) \frac{1}{2^k n}$

$= \cdots = 2^{-k}$

e.g. Chernoff + Markov

n balls into k buckets

prob twice average

$P[X/n \geq 2/k] \leq e^{-2n/k^2}$

(for a given bucket)

prob not a good hash

($\exists$ bucket with balls twice average)

$P[Y \geq 1] \leq \frac{E[Y]}{1}$

$= \frac{kE[Y_1]}{1} = ke^{-2n/k^2}$